

Data Scientist Takeaway Assessment

Objective: This assessment aims to evaluate your approach to handling real-world data challenges, your strategic vision for a data team, and your ability to translate technical details into business insights.

Task Overview

You are provided with a dataset. Perform the following:

- Exploratory Data Analysis: Identify patterns, outliers, and anomalies.
- Build a classification model to determine class of transaction.
- Insights and Recommendations: Provide insights from your analysis and model. How would these insights influence business decisions?
- Technical Documentation: Document your methodology, code, and findings in a clear and concise manner.

Dataset Name: sample_transactions_dataset.csv

Dataset Overview: The dataset represents financial transactions for a hypothetical fintech company over the course of 3 months. These transactions involve payments processed for various services and products.

Columns in the Dataset:

- Transaction_ID: A unique identifier for each transaction.
- Transaction_Date: date of the transaction (format: YYYY-MM-DD).
- Transaction_Time: The time the transaction was processed (HH:MM:SS).
- Customer_ID: A unique identifier for the customer.
- Amount: The amount of the transaction in USD.
- Status: The status of the transaction (Completed, Pending, Cancelled, Refunded, Chargeback).

Tasks for the Candidate:

1. Data Cleaning and Preprocessing:

- Handle missing values, outliers, and duplicate entries.
- Convert data types if necessary (e.g., converting string dates and times to datetime objects).

2. Exploratory Data Analysis:

- Analyze the distribution of transactions over time.
- Identify any patterns or trends that could indicate fraudulent activity or other anomalies.
- Analyze any other information that may be useful for the business.

3. Predictive Modeling:

- Build a classification/Regex model to predict the Classification_Tag of transactions.
- Evaluate the model's performance using appropriate metrics (accuracy, precision, recall, F1-score, ROC-AUC).
- Insights and Recommendations: Provide insights based on the analysis.

Data Privacy Note:

Please NOTE that all data used in this assessment is fictional and does not contain any real personal or financial information.

Submission Guidelines:

- Format: Submit your proposal, case study analysis, and scenario responses in a markdown document in a github repository. Source code in python to be submitted in the same github repository as well.
- Deadline: Please submit your completed assessment within 3 days of receipt.

Evaluation Criteria:

- Technical Proficiency: Ability to apply data techniques effectively and efficiently.
- Innovativeness: Creativity in problem-solving and in proposing innovative solutions.